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Brief papers

A multi-task learning framework for gas detection and concentration estimation

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ABSTRACT

Advancements in sensor devices have resulted in increased use of gas detection and concentration estimation in various fields. An electronic nose (E-nose) is an intelligent sensing device that mimics the human olfactory system and has been applied in gas analysis. We propose a multi-task learning–long short-term memory (MLSTM) recurrent network for simultaneous gas detection and concentration estimation, in which the two tasks share underlying features. The network utilizes the synergy between both two tasks, thereby enhancing the individual performances. Particle swarm optimization (PSO) proved a good method for optimizing the hyper-parameters used in the framework. The effectiveness of the framework and the parameter optimization method were verified by comparing the performances of the MLSTM model with those of eight models (two multi-task models, three classification models, and three regression models) using three datasets (two publicly available datasets and one dataset from the E-nose). The results showed that the proposed MLSTM model exhibited the best performance for simultaneous gas detection and concentration estimation because it was able to capture both global and local information.

1. Introduction

Gas is one of the four fundamental states of matter (the others being solid, liquid, and plasma) and is comprised of elements and molecular compounds, or mixtures of both [1]. Over the past decades, the advancement in sensor devices has made gas detection and concentration estimation increasingly popular in different fields, such as manufacturing [2], environmental control [3,4], food and beverage and science [5], and pharmaceutical [6]. Researchers have designed new smart devices for gas detection with novel sensor systems and pattern recognition systems. An electronic nose (E-nose) is a bionic olfactory device that is simple, effective, and low-cost system; the system consists of a gas sensor array integrated with an intelligent algorithm. An E-nose is a useful smart device for gas detection and has been used for gas detection and concentration estimation in conjunction with machine learning algorithms. Hassan et al. [7] proposed a binary decision tree approach for the classification of five different gases based on an E-nose. The algorithm achieved good performance data with the same concentration was used for training and testing, but the accuracy decreased significantly when using data with different concentrations. Zhang et al. [8] presented two approaches using a single multiple-input-multiple-output (SMIMO) and a multiple-input-single-output (MMISO)-based multilayer perceptron (MLP) with parameter optimization in a neural network for the detection of six kinds of indoor air contaminants. Although the MMISO based MLP exhibited better performance than SMIMO regarding the estimation accuracy and the time consumption of algorithm convergence, the method resulted in large errors. Chen et al. [9] developed a method for estimating the concentration of mixed volatile organic compounds (VOC) based on neural networks and decision tree learning. The relative error was less than 5% when the predicted concentration was higher than 20 ppm. However, the sensors were more sensitive to background interferences when detecting the gaseous mixture at low concentrations (0–20 ppm), which resulted in a higher relative error. Huang et al. [10] proposed a polynomial-based optimization method for the simultaneous estimation and classification of target gases using an E-nose system. Although the authors achieved a classification accuracy of 94% and good overall estimation performance, there existed several relatively large biases.

Multitask learning (MTL) is a method of inductive transfer that enhances the performance of one task by utilizing the domain in-

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formation contained in the training signals of other related tasks [11]. Min et al. [12] proposed a multimodal multi-task framework based on a deep belief network for recipe retrieval and exploration; both the recipe attributes and the multimodal content information were modeled. Wu et al. [13] presented a multitask strategy for quantitative toxicity analysis and prediction of small molecules. Cao et al. [14] developed a novel MTL formulation to explore the correlation between magnetic resonance imaging and cognitive measures. Collobert and Weston [15] defined a general network architecture and described its application to natural language processing (NLP) tasks including part-of-speech tagging, chunking, named-entity recognition, learning a language model, and the task of semantic role-labeling. Song et al. [16] applied MTL for automatic thyroid nodule detection and recognition. In addition, research has shown that learning multiple related tasks simultaneously always performs better than learning them independently [17]. Although the MTL framework has been successfully used in several areas, there are few reports in the field of gas identification. Inspired by previous successful studies, we propose an MTL framework based on a long short-term memory (LSTM) network [18,19] (MLSTM) and a recurrent neural network (RNN) composed of LSTM units, for simultaneous volatile classification and concentration estimation. In the framework, two shared layers are designed to learn common features of the tasks that are trained jointly with multiple loss functions. LSTM is commonly used in other fields, such as computer vision [20,21]. In this study, the LSTM is used as the backbone network, which means that we use the LSTM to extract essential features; this approach has been rarely used for the classification and concentration estimation of gas. In this way, the features can capture more valuable information on the different volatiles (gases), thereby resulting in improvements in the performances of the individual tasks.

In summary, the contributions of this study are as follows:

(1) The study is the first application of a multitask structure to improve the performance of simultaneous gas classification and concentration estimation based on the LSTM as a backbone network. The shared basic features, which are extracted by the LSTM layers, provide a better understanding of different aromas with different concentrations, thereby improving the performances of the individual tasks.
(2) Particle swarm optimization (PSO) is used to optimize the hyperparameters of the MTL structures. This optimization method provides higher accuracy and better applicability of the proposed method than other mainstream method. The proposed scheme has great potential for improving the performance of other machine learning-based olfaction tasks.

The rest of this paper is organized as follows. Section 2 presents an overview of related studies on LSTM networks. The MTL framework is detailed in Section3. Section 4 describes the experimental configurations and the results and discussion. The conclusions are provided in Section 5.

2. Related works

2.1. Long short-term memory (LSTM)

An LSTM is an RNN network that is able to learn long-term dependency information; it was proposed by Hochreiter and Schmidhuber in 1997 [18] and has been developed and improved by researchers in recent years.

As shown in Fig. 1, a common LSTM unit is composed of a cell, an input gate, an output gate, and a forget gate. The basic concept of the LSTM unit is to regulate the flow of information into and out of the cell by using different gates (input gate, forget gate, and output gate). In this way, LSTM units easily control the extent to which new information (values) flows into the cell, remains in the cell, and which value in the cell is used to compute the output activation of the LSTM unit. The mathematical description is as follows:

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci}c_{t-1} + b_i) \tag{1}$$

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf}c_{t-1} + b_f) \tag{2}$$

$$c_t = f_t c_{t-1} + i_t \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c) \tag{3}$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + W_{co}c_t + b_o) \tag{4}$$

$$h_t = o_t \tanh(c_t) \tag{5}$$

where i_t is the update information of the input gate at time t; similarly, f_t is the forgotten information of the forgotten gate, c_t is the state of the "memory cell", o_t is the output information of the output gate, h_t is the current neuron output; in addition, $\sigma(.)$ usually represents a sigmoid function.

2.2. Particle swarm optimization (PSO)

PSO is a global optimization method based on an evolutionary strategy and, is suitable for solving problems where the optimal solution is a point in a multidimensional space of the parameter [22]. PSO possesses several advantages over other heuristic optimization techniques, including a simple concept, easy implementation, robustness to control parameters, and computational efficiency [23]. The basic PSO algorithm for the calculation of the next position of the particle is [24]:

$$\begin{aligned} v_{i+1}(j) = v_i(j) + c_1 \times r_{1,i}(j) \times [pbest(j) - x_i(j)] \\ + c_2 \times r_{2,i}(j) \times [gbest(j) - x_i(j)] \end{aligned} \tag{6}$$

$$x_{i+1}(j) = v_{i+1}(j) + x_i(j) \tag{7}$$

where i is the number of iterations, j represents the j-th particle, $x_i(j)$ is the position (possible solution) of the j-th particle at the iteration i, $v_i(j)$ is the velocity of the particle j at the iteration i, c_1 and c_2 are positive constants, which are usually set equal to 2, $r_{1,\,i}$ and $r_{2,\,i}$ are random variables in the range (0, 1), *pbest(j)* is the j-th particle's local best position up to the iteration i, and *gbest(j)* is the value of the particle j of the best solution located by the swarm up to iteration i. Subsequently, several modified PSO algorithms were proposed, one of which incorporated an inertia weight *(w)* and was defined as:

$$\begin{aligned} v_{i+1}(j) = w \times v_i(j) + c_1 \times r_{1,i}(j) \times [pbest(j) - x_i(j)] \\ + c_2 \times r_{2,i}(j) \times [gbest(j) - x_i(j)] \end{aligned} \tag{8}$$

In this study, the parameter optimization process is based on Eqs. (7) and (8). In general [22], it has been shown that the algorithm worked well for most of the applications when $c_1 = c_2 = 2$. However, in this study, it was found that a value of 1.5 was optimal.

The inertia weights *(w)* are important for improving the performance of the PSO algorithm. Specifically, the selection strategy of the appropriate parameters w is crucial [25]. The strategy of the linear decreasing inertia weight (LDIW) is a simple and effective method for disposing on some low-dimension data. In our study, LDIW was used to adjust the weights using Eq. (9):

$$w(i) = w_{\max} - \frac{w_{\max} - w_{\min}}{i_{\max}} \times i \tag{9}$$

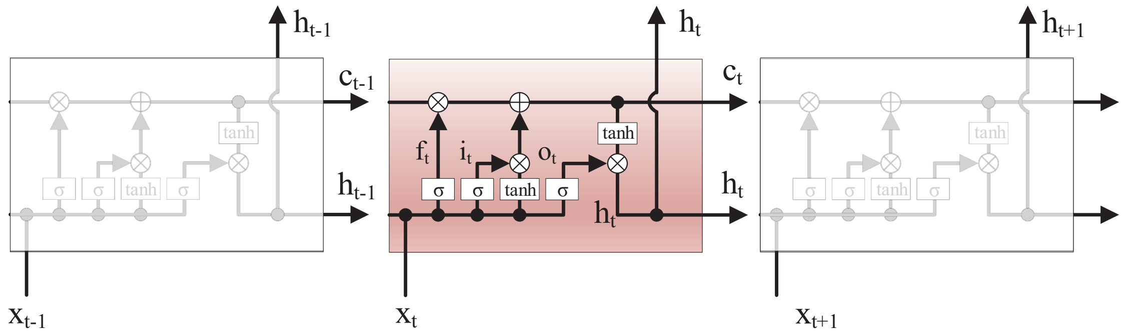


Fig. 1. Schematic diagram of the LSTM network structure.

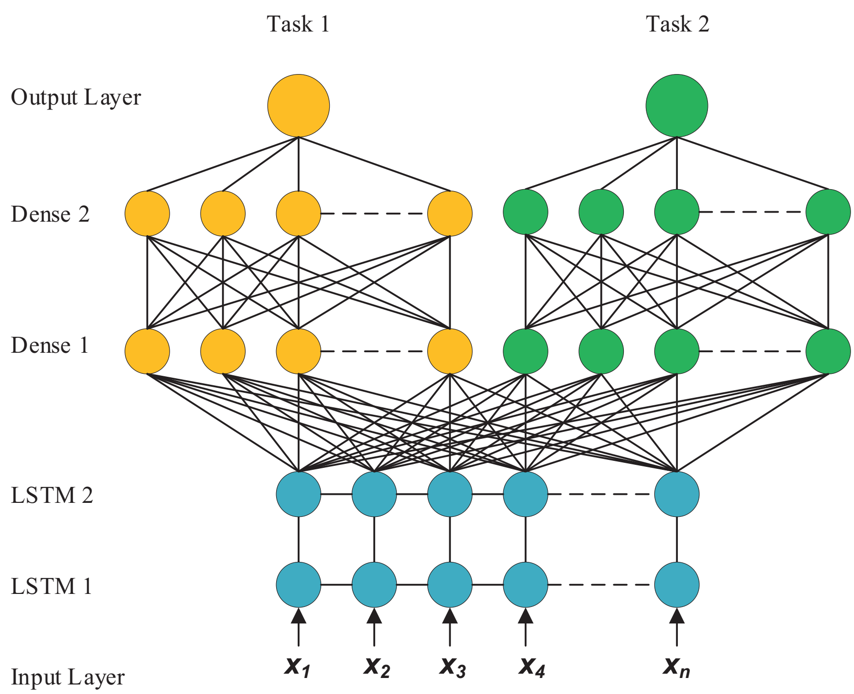


Fig. 2. The MTL framework based on the LSTM.

where, w_{max} and w_{min} are the maximum and minimum values of w, respectively; i and i_{max} are the current and maximum iterations, respectively.

Here, w is decreased from 0.9 to 0.3 using the LDIW to ensure that the PSO algorithm covers a larger area at the beginning and quickly locates the approximate location of the optimal solution.

3. Proposed method

The proposed multitask framework based on the LSTM (ML-STM) is shown in Fig. 2; the two task-specific feedforward subnets are stacked in parallel. The structure of the framework consists of two shared layers and two individual fully connected layers. Rectified linear units (ReLUs) are used to activate the neurons in the hidden layers, while the output units utilize the Softmax function and Sigmoid function for the outputs of the 1st task and 2nd task, respectively.

In this study, we have two tasks, as shown in Fig. 2; the training data for the tasks are the same and are denoted as $(\mathbf{x}_i, y_i^t)_{i=1}^{N_t}$, where $t = 1, 2$ (sequence number of tasks), $i = 1, \ldots, N_t$, N_t represents the number of samples of the t-th tasks, and $\mathbf{x}_i$ and y_i^t are the feature vectors that consist of the training features and target labels in the t-th task, respectively. The training goal of the proposed MTL is to minimize the following loss function for the two tasks simultaneously:

$$\text{argmin} \sum_{i=1}^{N_t} L(y_i^t, f^t(\mathrm{x}_i; \{\mathrm{W}^t, \mathrm{b}^t\})) \tag{10}$$

where f^t denotes a function of the feature vector $\mathbf{x}_i$ parameterized by a weight vector W_t and bias term b_t; L denotes the loss function.

In this model, the loss function for the 1st task (classification task) is the cross entropy loss, thus the loss of the 1st task is defined as:

$$L_1 = -\sum_{i=1}^{N_1} y_i^1 \log(f^1(\mathrm{x}_i; \{\mathrm{W}^1, \mathrm{b}^1\})) \tag{11}$$

For the 2nd task (regression task), the mean squared error of the loss is usually used; thus the loss function is defined as:

$$L_2 = \frac{1}{2}\sum_{i=1}^{N_2} (y_i^2 - f^2(\mathrm{x}_i; \{\mathrm{W}^2, \mathrm{b}^2\}))^2 \tag{12}$$

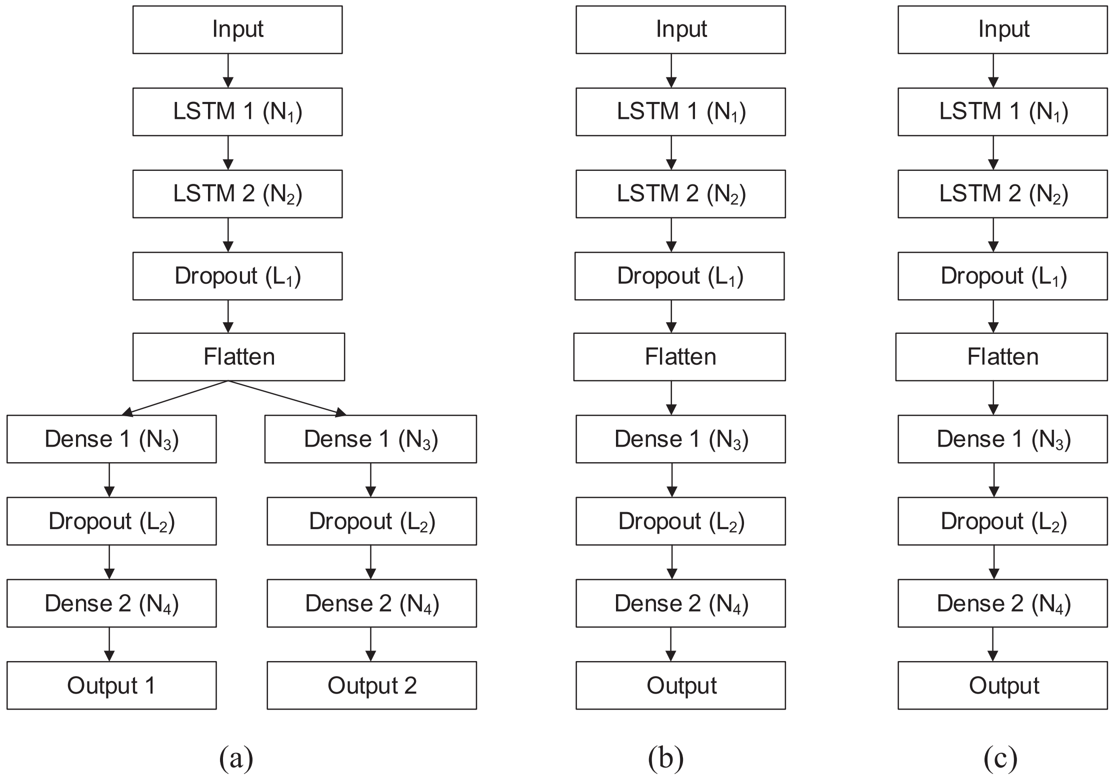


Fig. 3. LSTM-based architectures for (a) simultaneous gas detection and concentration estimation, (b) gas detection, and (c) concentration estimation. The N_X in brackets on the right denote the cell size, or number of neurons, and the L_X denote drop rate in the LSTM layer, dense layer, and dropout layer, respectively.

To minimize overfitting, it is usually advisable to add a regularization term for the weight vectors, which provides an advanced loss function for the task:

$$L = \alpha L_1 + (1-\alpha)L_2 + \beta\left(\left\|W^1\right\|_2^2 + \left\|W^2\right\|_2^2\right) \tag{13}$$

where $\|\bullet\|_2$ represents the L2 norm, α is the weighting factor that ranges from 0.0 to 1.0, β is the penalty constant of L1 and L2, respectively. All the hyper-parameters are optimized by the PSO algorithm.

4. Experiments

4.1. Network architectures

To emphasize the importance of the multitask approach and fusion of the bottom layers of the LSTM, we evaluate three LSTM-based architectures, as shown in Fig. 3; one is for both the gas detection and concentration estimation and the others are for the individual tasks of gas detection and concentration estimation. The three models are named MLSTM, classification based learning–long short-term memory (CLSTM), and regressionbased learning–long short-term memory (RLSTM), respectively.

Random forest (RF), support vector machine (SVM), and back propagation neural network (BPNN) are machine learning algorithms that have been used for many tasks in different fields; the input for these algorithms is a one-dimensional feature vector. RF is applicable to classification problems with unbalanced, multiclass and small sample size data without data preprocessing [26]. SVM is suitable for small data sets for classification tasks [27]. BPNN is a popular method due to its ability to handle nonlinear data.

Fernández–Delgado et al. [28] evaluated 179 classifiers using 121 data sets, which represented the entire University of California-Irvine (UCI) data base (excluding large-scale problems) and other real problems. It is worth noting that in the 121 data sets, the samples ranged from tens to tens of thousands, the number of features ranged from three to ten (all features were one-dimensional vectors), and the categories were not the same. The three highest performing models were RF, SVM, and BPNN.

In our experiments, we used three data sets (two from the UCI, and one from the authors' lab) whose structure information were similar to that of the data sets used by Fernández–Delgado et al. We compared the performance of the MLSTM framework and the PSO method with several other methods: (1) a multi-task framework based on the BPNN (MBPNN); (2) we compared the performance of the LSTM (CLSTM, and LSTM), which was used for the classification task or regression task; (3) RF and SVM were used to compare the classification task or regression task with the LSTM.

4.2. Experiments using three datasets

4.2.1. Experiment A (dataset I: twin gas sensor array dataset)

This dataset was obtained from the UC Irvine Machine Learning Repository [29] and includes the recordings of five replicates of an 8-sensor array. Each unit holds 8 metal oxide (MOX) sensors and has integrated custom-designed electronics for the sensors operating the temperature control and signal acquisition. Each day, one of the five measurement units was exposed to different gas conditions (ethanol, methane, ethylene, and carbon monoxide at 10 concentration levels each) to collect data in random order. The experiment lasted 16 days, and 640 samples were tested. The duration of each experiment was 600 s (the preliminary stabilization phase was 50 s, the testing phase was 100 s, and the cleaning phase was 450 s) and the conductivity of each sensor was acquired at 100 Hz [30].

In experiment A, we used 448 samples (70%) as training samples (training set) and the remaining 192 samples (30%) were testing samples (testing set). Also, we used measurements of each

Table 1
The best parameter combinations based on MLSTM with different swarm sizes.

Swarm Size	Parameters								Classfication		Regression	
	N_1	N_2	N_3	N_4	L_1	L_2	α	β	Acc.	MAE	r	R^2
10	27	34	32	38	0.1	0.2	0.5	0.01	0.9910	21.64	0.9298	0.7942
15	22	37	23	42	0.1	0.1	0.4	0.02	0.9836	20.26	0.9061	0.8052
20	28	32	29	33	0.1	0.1	0.4	0.01	0.9801	22.89	0.9189	0.7594
25	29	31	30	38	0.2	0.2	0.5	0.01	0.9835	21.26	0.9190	0.8095
30	28	29	35	36	0.1	0.2	0.4	0.02	0.9921	19.90	0.9158	0.8162
35	26	36	31	35	0.2	0.1	0.5	0.02	0.9908	19.81	0.9266	0.8286
40	25	33	24	40	0.1	0.1	0.4	0.02	0.9855	25.02	0.8967	0.7489
45	28	30	27	36	0.2	0.1	0.5	0.01	0.9921	10.41	0.9854	0.9172
50	26	32	29	39	0.2	0.2	0.4	0.02	0.9999	4.038	0.9985	0.9944
55	27	35	32	36	0.2	0.2	0.4	0.02	0.9965	7.721	0.9889	0.9775
60	29	35	28	37	0.2	0.1	0.4	0.02	0.9899	21.13	0.8966	0.7990

Table 2
Parameter list of the models.

Parameters	Swarm size	N_1	N_2	N_3	N_4	L_1	L_2	α	β
MLSTM	50	26	32	29	39	0.2	0.2	0.4	0.02
MBPNN	55	28	38	30	35	0.2	0.2	0.4	0.01
CLSTM	50	25	37	31	37	0.2	0.2	1.0	0.02
RLSTM	50	26	34	29	38	0.2	0.2	0.0	0.02

sample in the last 10 s as input features which were regarded as stable features for model training and testing. Therefore, the two datasets were created and were expressed as a 448,000 × 8 matrix for training and a 192,000 × 8 matrix for testing, respectively.

Each feature was scaled to cover a range between zero and one. During training, three-fold cross-validation was applied to assess the training performance of the LSTM-based models. During training, the dropout strategy was used, which involved randomly setting a fraction rate of input units to zero at each update to avoid overfitting. This method was conducted via PC programming using the Python language based on two open-source machine learning platforms (Keras and TensorFlow).

The hyper-parameters were optimized by PSO during model training, in which the networks' total loss was used as the fitness function in the PSO. For the four neural network-based architectures, there were eight hyper-parameters, namely, the dimensionality of the output space of each layer, the dropout rate (fractions of the LSTM 2 and Dense 2 input units to drop, which were named L_1 and L_2, respectively), α, and β, which were optimized.

As shown in Table 1, the eight parameters were considered as the eight dimensions of the particles' positions. The positions of the particles were initialized randomly: N_1, N_2, N_3, $N_4 \in [10, 50]$, L_1, $L_2 \in \{0.1, 0.2\}$, $\alpha \in [0, 1.0]$, $\beta \in \{0.01, 0.02\}$ according to our practical experience. The swarm size usually affects the performance of the PSO. Therefore, we conducted 11 experiments using different swarm sizes ranging from 10 to 60 with a step size of 5 and observed the effect of the swarm size on the experimental results. The hyper-parameter optimization of the MLSTM model is used as an example, Table 1 shows the best combination of parameters optimized by the PSO with different swarm sizes and the experimental results of the MLSTM model with the best parameter combination. The final swarm sizes were set to 50 in this experiment.

In the subsequent experiments, the parameter optimization processes of another three methods (MBPNN, CLSTM, and RLSTM) were the same as those of the MLSTM methods, and the details are not described again. The optimal hyper-parameters of the four networks in experiment A are shown in Table 2.

The model performance for the dataset I was evaluated by determining the classification accuracy. In addition, the mean absolute error (MAE), correlation coefficient (Pearson product-moment correlation coefficient (r)), and the coefficient of determination (R^2) were used to evaluate the regression performance. The mathematical expressions of the parameters are shown in Eqs. (14)–(16), where y_i is the label of the i th sample, $\widehat{y_i}$ is the predicted value of the i th sample, and the horizontal line of the superscript indicates the average value.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|\widehat{y_i} - y_i| \tag{14}$$

$$r = \frac{\sum_{i=1}^{n}(y_i - \bar{y})(\widehat{y_i} - \bar{\widehat{y}})}{\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}\sqrt{\sum_{i=1}^{n}(\widehat{y_i} - \bar{\widehat{y}})^2}} \tag{15}$$

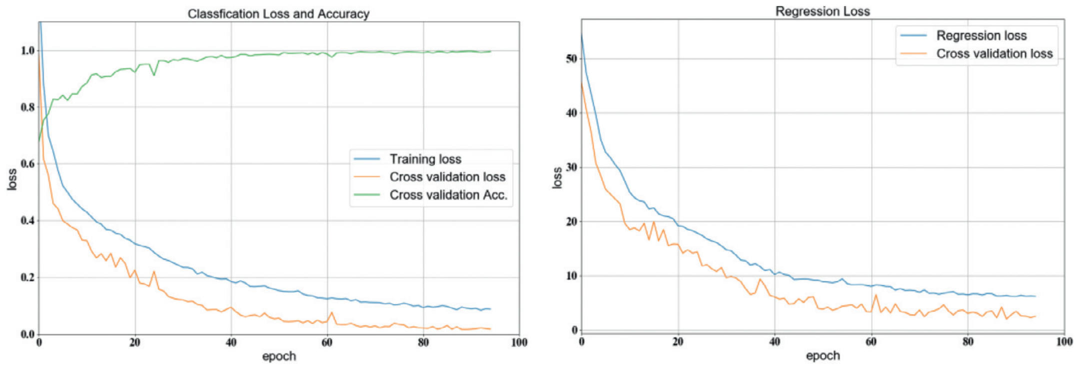
$$R^2 = 1 - \frac{\sum_i^n (y_i - \widehat{y_i})^2}{\sum_i^n (y_i - \bar{y})^2} \tag{16}$$

The MAE was chosen because it is a straightforward index when the data have the same scale [31]. The R^2 is the proportion of the variance in the dependent variable that is predictable by the independent variable(s) [32] and r describes the correlation and dependence between two or more variables. Therefore, the higher the r and R^2 values and the lower the MAE, the better the fit of the regression model is.

As shown in Fig. 4, after 95 epochs of model training in the MLSTM architecture, the total loss falls to the preset threshold; the classification loss and regression loss curves are shown in Fig. 4(a) and (b), respectively.

The plots of the actual concentrations and predicted concentrations obtained from the MLSTM model for the test samples (200 randomly chosen test samples) is shownin Fig. 5, where the y-axis represents the concentration and the x-axis denotes the sample number. The results show a good agreement between the predicted and actual values for the MLSTM model. It was found that the MTL architecture was useful for predicting the gas concentration.

Table 3 summarizes the statistics of the experimental results for the eight models (two multi-task models, three classification-based models, and three regression-based models). The results show that the MLSTM model had the best performance for the individual classification and regression tasks, as well as for the simultaneous classification and regressiontasks.



(a) (b)

Fig. 4. Loss curves of the MLSTM in the training process for dataset I: (a) classification task, (b) regression task.

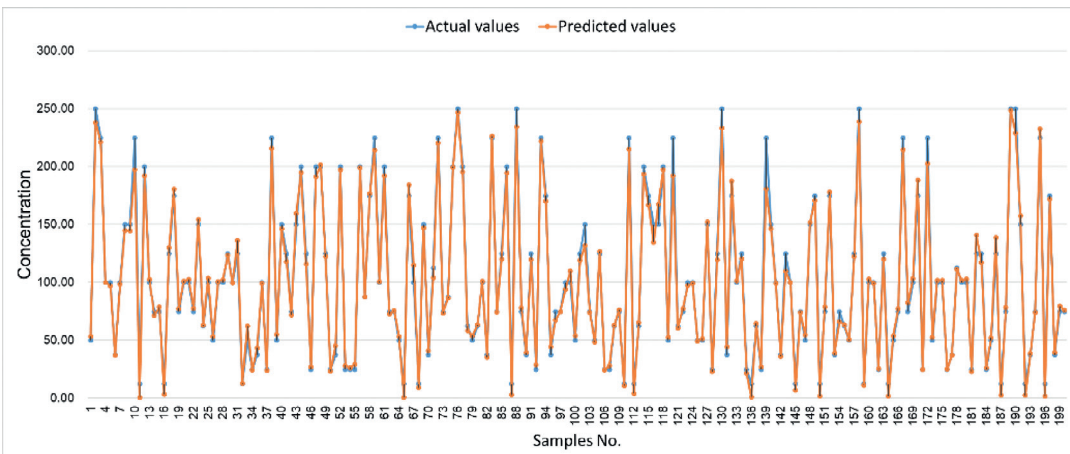


Fig. 5. The plots of actual values and predicted values of the test set.

Table 3
The performance of eight models for dataset I.

Classification task		Regression task		
	Acc.	MAE	r	R^2
MLSTM	0.9999	4.038	0.9985	0.9944
MBPNN	0.9892	19.54	0.9187	0.8071
CLSTM	0.9903	–	–	–
RLSTM	–	17.84	0.9395	0.8448
SVM	0.9928	–	–	–
SVR	–	23.93	0.8810	0.6703
RF	0.9915	–	–	–
RFR	–	14.68	0.9255	0.8526

4.2.2. Experiment B (dataset II: gas sensor drift dataset and experiments)

A publicly available data set created by Vergara et al. [33,34] was also obtained from the UC Irvine Machine Learning Repository and used in our experiments to evaluate the performance of the proposed model for gas classification with sensor drift. The dataset was gathered over 36 months and contained 13,910 measurements from 16 MOX gas sensors exposed to six gases at different concentration levels. Samples were obtained from all possible gas type/concentration pairs and sampling was conducted randomly. Each sample included a class label, concentration, and a 128-dimensional feature vector (8 features (two steady-state features and six transient features) × 16 sensors) describing the steady-state and the dynamic-state gas sensing responses.

As shown in Table 4, the samples were split into 10 batches according to the acquisition time to ensure a relatively uniform distribution of the number of experiments in each batch (the six gases ethanol, ethylene, ammonia, acetaldehyde, acetone, and toluene are abbreviated as etha, ethy, ammo, acetal, acet, and tolu). The objective of this experiment was to classify the gases and predict their concentrations simultaneously.

We used the samples in batch 7 as training samples and those in batch 6 as testing samples to ensure that sufficient information could be obtained from the samples for learning. This evaluation strategy was suitable to verify the performance of the model. In the dataset, each sample is represented by 128 features extracted from the sensors' response curves. Each feature was first normalized to zero mean and unit variance in each batch. Considering that the sample concentrations ranged from tens to thousands of ppm, the concentration value of each sample was used as the label for the regression. During training, 3-fold cross-validation was used to evaluate the performance of the three architectures and the final evaluation was executed on the testing set. During training, a dropout strategy was also used to prevent overfitting.

Table 5 shows the eight hyper-parameters of the four architectures. The parameters were optimized using the PSO, which resulted in an excellent performance in this experiment.

The evaluation criteria of all the methods in this experiment were the same as those in experiment A. Table 6 summarizes the statistical parameters of the experimental results for the different architectures. The MLSTM exhibited the best performance. The experiment proved that the MLSTM was not only capable of handling

Table 4
The details of dataset II.

Batch No.	Months	Etha	Ethy	Ammo	Acetal	Acet	Tolu	Total
1	1–2	90	98	83	30	70	74	445
2	3–4, 8–10	164	334	100	109	532	5	1244
3	11–13	365	490	216	240	275	0	1586
4	14–15	64	43	12	30	21	0	161
5	16	28	40	20	46	63	0	197
6	17–20	514	574	110	29	606	467	2300
7	21	649	662	360	744	630	568	3613
8	22–23	30	30	40	33	143	18	294
9	24, 30	61	55	100	75	78	101	470
10	36	600	600	600	600	600	600	3600

Table 5
Parameter list for the models with optimum performance.

Parameters	Swarm size	N_1	N_2	N_3	N_4	L_1	L_2	α	β
MLSTM	45	129	156	48	54	0.2	0.2	0.4	0.01
MBPNN	55	135	148	49	51	0.2	0.1	0.4	0.01
CLSTM	50	130	151	41	57	0.1	0.1	1.0	0.02
RLSTM	45	126	142	50	55	0.1	0.1	0.0	0.02

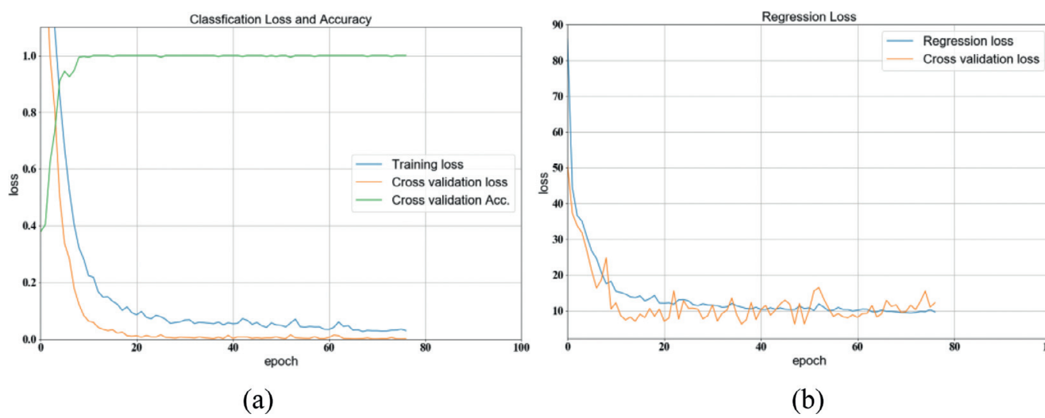


Fig. 6. Loss curves of the MLSTM in the training process for Batch 7 of dataset II: (a) classification task, (b) regression task.

Table 6
The performance of the eight models for dataset II.

	Classification task	Regression task		
	Acc.	MAE	r	R^2
MLSTM	0.9517	14.949	0.9486	0.9073
MBPNN	0.7648	45.356	0.8336	0.4906
CLSTM	0.8478	–	–	–
RLSTM	–	25.708	0.8764	0.7472
SVM	0.9665			
SVR		23.564	0.8965	0.7887
RF	0.9487			
RFR		24.860	0.9039	0.7348

multiple tasks simultaneously but also has good generalization performance when the appropriate parameters were selected. This experiment demonstrated that learning correlated tasks simultaneously boosted the performance of the individual tasks.

The loss curves of the MLSTM architecture in the training process are shown in Fig. 6. After 77 epochs of model training, the total loss falls to the preset threshold; the classification loss and regression loss curves are shown in Fig. 6(a) and (b), respectively.

The plots of the actual concentrations and the predicted concentrations obtained from the MLSTM model is shown in Fig. 7, where the y-axis represents the concentration and the x-axis denotes the sample number. The plots also show the regression results of the concentration, indicating that the MTL architecture was effective for predicting the gas concentration, even though the training samples and testing samples were from different batches. This result demonstrates that the proposed MLSTM model has good generalization performance.

4.2.3. Experiment C (dataset III: custom dataset from the authors' lab)

We used another dataset of volatiles from two solutions with different concentrations to assess the model performance in a real scenario. The dataset was obtained by the MOS-E-nose (PEN3 E-nose, Airsense Analytics GmbH, Germany). The details of Metal-Oxide-Semiconductor (MOS) sensors in the E-nose are shown in Table 7.

In this experiment, the data for the gas detection consisted of two types of solutions (marked as A and B), which were prepared in the authors' lab. Table 8 shows the details of the dataset; three samples of each concentration (of A and B) were prepared and measured daily for 10 days. A total of 420 samples were tested. All experiments were performed in the authors' laboratory at a temperature of 25 ± 1 °C and relative humidity of 50 ± 2%.

Similarly, 294 samples (70%) were used as training samples (training set) and the remaining 126 samples (30%) were the testing samples (testing set).

For the dataset obtained in our lab, we adjusted the domain of the hyper-parameters. The best combinations of parameters of the four network-based models are shown in Table 9. The PSO algorithm was used for optimization.

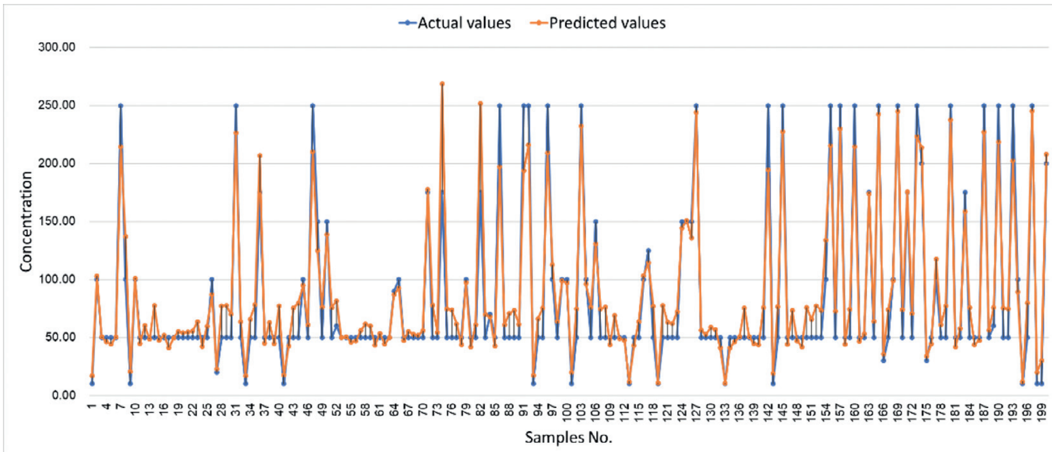


Fig. 7. The plots of the actual values and predicted values of Batch 6.

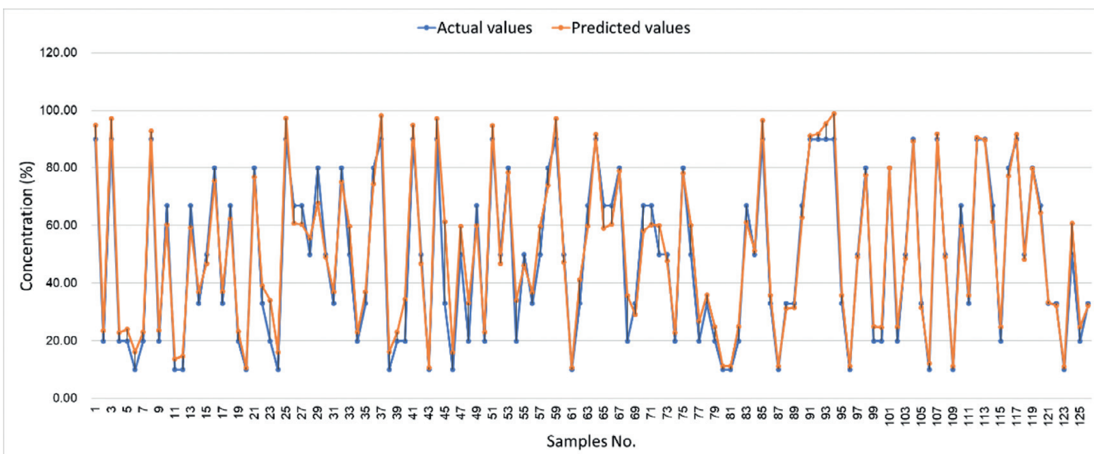


Fig. 8. The plots of actual values and predicted values of the testing set.

Table 7
Standard sensor array in the PEN3 E-nose.

Number	Sensor	Sensed substances
MOS1	W1C	aroma constituent
MOS2	W5S	sensitive to nitride oxides
MOS3	W3C	ammonia, aroma constituent
MOS4	W6S	hydrogen
MOS5	W5C	alkane, aroma constituent
MOS6	W1S	sensitive to methane
MOS7	W1W	sensitive to sulfide
MOS8	W2S	sensitive to alcohol
MOS9	W2W	aroma constituent, organic sulfur compounds
MOS10	W3S	sensitive to alkane

Table 8
The details of dataset III.

Type No.	Name of solution	Concentrations (%)						
A	ethanol	10	20	33	50	67	80	90
B	n-propanol	10	20	33	50	67	80	90

Table 10
The performance of the eight models for dataset III.

	Classification task	Regression task		
	Acc.	MAE	r	R^2
MLSTM	1.0	4.8269	0.9753	0.9430
MBPNN	1.0	13.2736	0.8782	0.7703
CLSTM	1.0	–	–	–
RLSTM	–	5.3986	0.9564	0.9145
SVM	1.0			
SVR		13.1355	0.9473	0.5923
RF	1.0			
RFR		7.3350	0.9230	0.8482

Table 10 summarizes the statistics of the experimental results for the different architectures. Although the MLSTM and MBPNN achieved 100% accuracy for the identification of the two solutions, the performance of MLSTM is better than that of MBPNN in the regression tasks. The three classification models (CLSTM, RF, and

Table 9
Parameter list for the models with optimum performance.

Parameters	Swarm size	N_1	N_2	N_3	N_4	L_1	L_2	α	β
MLSTM	40	21	38	22	37	0.2	0.2	0.4	0.01
MBPNN	45	24	40	25	34	0.2	0.1	0.4	0.01
CLSTM	35	20	35	31	47	0.1	0.1	1.0	0.02
RLSTM	40	22	34	35	40	0.1	0.1	0.0	0.02

SVM) and three regression models (RLSTM, RFR, and SVR) had good performance in this experiment because of the samll sample size. The proposed MLSTM model had the best performance, indicating that the model is well suited for pratical scenarios.

The plots of the actual concentrations and the predicted concentrations obtained from the MLSTM model is shown in Fig. 8, where the y-axis represents the concentration and the x-axis denotes the samples number. The plots also depicts the regression results. The good agreement between the actual concentrations and the predicted concentrations indicate that the MTL architecture was effective for predicting the concentration of the solutions.

5. Conclusion

In this study, we proposed a multitask framework for simultaneous gas detection and concentration estimation using the MLSTM model and conducted experiments. The MTL framework exhibited excellent performance, which was attributed to the network's ability to learn more discriminative features. The fusion of the bottom layers enhanced the performance for the structure-dependent tasks of gas detection and concentration estimation because the features were invariant to the geometry in the deeper layers of the LSTM. Our proposed method utilized these characteristics to enhance the performance of gas detection and concentration estimation.

The proposed MLSTM method is based on MTL and LSTM with a deep architecture of the LSTM recurrent network with two task branches that share the underlying features. The experimental results showed that the model accurately predicted the gas category and its concentration, regardless of whether the testing and training samples were from the same batch (measured in the same time period), which indicated that the model had good generalization performance. In addition, the proposed deep model was able to describe the nonlinear relationship between the system inputs and outputs, particularly if the sensor data were heterogeneous, complex, and had missing data points.

The quantitative results of the experiments indicate that the proposed method can perform the tasks of gas detection and concentration estimation simultaneously.

This study provides an in-depth analysis of the use of MTL for a new application (gas detection and concentration estimation). The results of the study can assist manufacturers in standardizing operating processes and reducing costs while promoting the practical use of the equipment (E-nose). Future work is underway to adapt the proposed method to other applications such as the assessment of food quality and safety.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Huixiang Liu: Conceptualization, Methodology, Software, Data curation, Writing - original draft. **Qing Li:** Validation, Formal analysis, Writing - review & editing. **Yu Gu:** Investigation, Resources, Writing - review & editing, Supervision, Project administration, Funding acquisition.

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